Sorting · my notes L08

a Comparator = a list of keys, each with a direction

sorted()

natural order · needs **Comparable**

String, numbers, dates, BigDecimal, enums ✓

a record → ClassCastException 💥 (at run time!)

sorted(Comparator)

any type, any order

`comparing(Policy::premium)`

→ POL-505 185.00, POL-509 210.00 …

The podium 🏆 tier first, then total, then name

C010 has more money than C006, but the tier key comes FIRST



`comparing(tier, reverseOrder())` → `.thenComparing(total, reverseOrder())` → `.thenComparing(name)`

reversed() = flip the WHOLE ladder 🔄

(tier, name).reversed()

tier DESC ✓

names DESC too ✗

C006 C003 | C010 C007 …

comparing(tier, reverseOrder())

tier DESC ✓

names ASC ✓

C003 C006 | C001 C004 …

reversed() also moves `nullsLast` → nulls first!

Toolbox 🧰

- `comparingInt(Flight::seats)` no boxing
- `comparingDouble(load)` ratios only
- `naturalOrder()` / `reverseOrder()`
- enums: SILVER < GOLD < PLATINUM (declaration order)

Exercise: departures ✈

date → time (TBA last) → code

TR808 07:15, SQ12 09:10, GA827, MH602, SQ318 TBA, UL309 TBA, SQ938, SQ21

Nulls, case and stability

null key 💥

ACC-1006, ACC-1015: no transactions → last = null

`comparing(last)` → NPE

`comparing(last, nullsLast(…))` ✓

Aa vs aA

natural: [CARD, PayPal, card, paynow] (uppercase first: Unicode)

`CASE_INSENSITIVE_ORDER` : [CARD, card, paynow, PayPal]

Stable = equal keys keep their place in the queue 🚶 🚶

August by date, newest first: … 26: `T012` , 26: `T013` … 22: `T009` , 22: `T010` … (input order kept)

Map by value 🗺

TreeMap (by KEY)

ACC-1001=4, ACC-1002=2, ACC-1003=7 …

→

```
entrySet().stream()
.sorted(comparingByValue(reverseOrder()))
.limit(3)
```

→ [ACC-1003=7, ACC-1004=6, ACC-1008=6]

list.sort

in place · List.of → UnsupportedOperationException

stream().sorted()

new list · source untouched (SQ318 still first)

reversed() [21]

a view of the other end · bottom 3 → C009, C002, C005

★ Remember

keys, not compare() · reverseOrder() per DESC key · reversed() flips everything · nullsLast for null keys · enums = declaration order · stable + tie-breaker · Exercise: departures board with TBA · next → L09 terminal operations & reduce